Classical Object Detectors vs. MLLMs for Child Detection in Images: A Preliminary Study

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Abstract

This paper presents the Parenting Slop Dataset, a video dataset collected from a social media platform with keyframe-level annotations indicating the locations of children in multimedia content. Alongside dataset development, three approaches to child detection are investigated. A key constraint is that none of the employed models are trained or fine-tuned, eliminating the need to collect additional training data. The first approach combines classical object detectors, such as YOLO and Faster R-CNN, with age estimation. The second uses multimodal large language models (MLLMs) for direct child localization, while the third combines classical object detection with MLLM-based age estimation. The study examines their applicability, advantages, and limitations. The results show that classical detectors provide fast and precise person detection but are less flexible when the target objects do not match predefined categories. In child detection, age estimation can become a bottleneck, particularly when reliable age-related visual information is unavailable. MLLMs offer greater flexibility, but their outputs are less predictable and more difficult to control. The largest evaluated models outperform classical detectors in the target task and can provide contextual information that complements object localization. The results also indicate that combining classical detection with MLLM-based age estimation can provide an effective solution while maintaining a modular structure that can be extended to a more complex detection system.

1. Introduction

Child detection has applications across a range of computer vision tasks, including NSFW (not safe for work) and CSAM (child sexual abuse material) content identification, automatic image and video anonymization prior to publication, and the detection of synthetic content, including so-called AI slop targeting minors. Child detection can be regarded as a specialized subtask of general person and object detection. Researchers have extensively studied these broader tasks, and numerous high-performing models are available. However, these general-purpose solutions cannot be applied directly to child detection without modifications. A typical modification is to fine-tune the object detector on a dataset of images of children prepared for this task. Because collecting and annotating suitable data is highly time-consuming, a key assumption of this project is to minimize or eliminate the need for additional data collection whenever possible and instead investigate the applicability of existing machine-learning methods. Another constraint is that cloud-based infrastructure cannot be used to compute results, as the dataset may contain sensitive data in the future.

The primary objective of the project[1] is therefore to develop an effective approach for identifying minors in images and video footage by leveraging currently available models.

2. Parenting Slop Dataset

The story behind this dataset started when the author was on maternity leave. She noticed that social platforms began suggesting her more videos with children, including many synthetically generated clips of different animals stealing or attacking babies. She decided to download them, as more and more strange animals appeared in the clips. It was a starting point for the dataset collection for further analysis of children's presence in real and AI-generated videos, and it also explains the bias toward clips with animals and children.

The Parenting Slop Dataset[2] consists of 20 movie clips showing children in various situations. All of them were downloaded from the Facebook Reels platform. The dataset

[1] The code for the experiments is available on the GitHub platform: https://github.com/weronikagutfeter/MinorsReport

[2] The dataset can be downloaded from the platform Zenodo: https://zenodo.org/records/22092731



Figure 1. Sample frames from the Parenting Slop Dataset

contains both real and synthetic data. The clips were selected based on the presence of minors in different age groups and settings. The minor category is defined as a child with an apparent age below fourteen years. However, the age threshold requires further discussion as the definition of minor may vary by country and by the goals of the classification system. Fourteen years of age can be considered the minimum threshold below which multimedia content should, in most cases, be treated as sensitive. At present, the dataset does not include borderline cases, such as young adults or older adolescents, which may make the distinction between minors and adults difficult. Selected videos contain subjects in close-up and from a distance; children can be in groups, occluded, moving, and not looking at the camera, making the data more challenging.

All the videos were trimmed to 10 seconds to have a uniform length. They were then sampled at 5 fps and saved as JPG images, resulting in 50 frames per video, for a total of 1000 frames. Keyframes were annotated with rectangular bounding boxes for the *minor* category using LabelMe. Adults and other objects in the images are not labeled. Bounding boxes and annotations follow the COCO object detection format [1]. The annotations are provided only for the keyframes, but with appropriate approximation, they can also be used to evaluate algorithms on the corresponding source videos.

3. Methodology

The main goal of this experiment is to identify all children appearing in the images from the Parenting Slop Dataset. Although the dataset is inherently video-based, the present study treats it as a collection of static frames. The videos are sampled into individual frames, and all evaluated methods process these images independently. Approaches that exploit temporal or other dynamic information are therefore beyond the scope of this study.

Two common approaches to child detection in images can be distinguished. The first is to train a dedicated detector capable of identifying all subjects that meet the defined criteria of age. The second is to combine a general-purpose person detector with an age estimation method.

A dedicated child detector requires a suitable training dataset, which can be time-consuming and challenging to obtain. In particular, collecting images containing children may be subject to significant legal and ethical constraints. The detector's quality depends on the quality of the collected data, but after training, it provides a simple one-step inference solution. Dedicated child detectors based on YOLO have been proposed, for example, by the authors of [2]. However, as stated earlier, methods like these, which require additional training or fine-tuning, are excluded from the comparison.

Combining person detection and age estimation is a second common approach. Person and object detection are extensively studied tasks, with numerous well-established models available, including YOLO, Faster R-CNN, and RetinaNet. Traditional object detectors typically use convolutional neural networks or Transformer-based architectures and can accurately localize objects belonging to a predefined set of categories. The authors of [3], for example, proposed a pipeline based on the YOLOX detector and custom age-estimation methods based on different sets of subjects' characteristics. To identify minors in an image using this approach, the system first localizes all persons, then estimates the age of each detected person. This second step lets the system filter out detections classified as adults. However, state-of-the-art age estimation methods commonly rely on facial information, which can be problematic in real-world scenarios. In the Parenting Slop Dataset, faces may be small, partially occluded, blurred, or completely absent. These limitations can reduce the effectiveness of the age estimation stage and, consequently, the performance of the entire detection pipeline.

Multimodal large language models (MLLMs) offer an alternative and have recently been applied to a broad range of computer vision tasks. MLLMs can process and generate non-textual content, including images, video, and audio, and are trained to perform various vision-language tasks such as image classification, object localization, semantic segmentation and visual grounding. Their ability to jointly interpret visual content and natural-language instructions makes them a potential solution for end-to-end child detection. In this setting, a model can be prompted to identify and localize all children visible in an image without explicitly separating person detection and age estimation into two

stages. For comparison, a two-stage prompting strategy is also considered, in which the model is asked to identify persons and label them as adult or child, and this information is then used to filter part of the results.

Another particularly relevant family of methods consists of open-world object detectors, which combine conventional detection capabilities with text-based encoders to identify objects outside the predefined set of categories used during training. One example is Open-World YOLO [4]. Open-World YOLO incorporates an additional text encoder into the detection pipeline and uses contrastive learning to enable detection of objects not known during training. Zero-shot detection can also be performed using SAM (Segment Anything Model), where the third generation of the algorithm enables segmentation with a text prompt to a DETR-based detector [5]. In general, a combination of closed-set object detectors with the more flexible capabilities of MLLMs can be distinguished as the third approach to solving the problem defined in this work. In the hybrid solution proposed in this study, the MLLM module replaces age estimation methods for persons localized by a classical object detector.

Based on these considerations, the following baseline approaches for child detection can be defined:

1. **Classical person detector + age estimation:** A conventional object detector first identifies all people in the image. An age estimation model then classifies the detected regions as children or adults and filters out the latter.
2. **MLLM-based child localization:** An MLLM is prompted to identify and localize all children visible in the image, providing a direct end-to-end approach to child detection.
3. **Hybrid solutions:** Combining classical detectors and MLLMs

4. Experiments

As mentioned previously, the project excludes the use of external infrastructure for running experiments. Therefore, for both classical object detectors and MLLM-based methods, only open-source and open-weight models that could be executed locally on a personal computer were considered. All experiments were conducted on a laptop equipped with an NVIDIA GeForce RTX 3080 GPU and 32 GB of RAM.

Inference time is reported in some of the tables; however, the implementations and supporting libraries were not thoroughly optimized. Therefore, the reported execution times should be treated as approximate and should not be considered a fully fair comparison of computational efficiency between the evaluated methods.

For large language models, the selection of models was limited by the ability to successfully deploy the models on a local machine with the Llama.cpp framework. As a result, not all evaluated models could be selected to have comparable size and parameter count.

The detection results were evaluated using the standard metrics defined by the COCO [1] evaluation protocol. These include Average Precision computed across IoU thresholds ranging from 0.50 to 0.95 with a step of 0.05 (AP), Average Precision at an IoU threshold of 0.50 (AP-50), Average Precision at an IoU threshold of 0.75 (AP-75), and Average Recall with a maximum of 100 detections per image (AR).

5. Person detectors with age estimation

State-of-the-art (SOTA) object detectors are commonly trained and evaluated on the Common Objects in Context dataset (COCO) [1], which defines a fixed set of object categories. One of the categories included in COCO is person. Although this category does not distinguish between adults and children, it provides a basis for child detection. A straightforward way to adapt a general-purpose person detector to the task considered in this work is therefore to first detect all people in an image and subsequently determine their age.

Modern object detectors are based primarily on two architectural paradigms: convolutional neural networks (CNNs) and Transformers, although many recent models combine components from both approaches. Among the established CNN-based approaches, Faster R-CNN [6] is a representative two-stage detector. It first generates region proposals using a Region Proposal Network (RPN) and then classifies these regions and refines their bounding boxes. Mask R-CNN extends Faster R-CNN by adding a parallel mask prediction branch, allowing the model to perform instance segmentation in addition to object detection. Cascade Mask R-CNN further extends this concept by using a sequence of detection stages with progressively higher Intersection over Union (IoU) thresholds. Another family of object detectors is YOLO (You Only Look Once), which represents the single-stage detection paradigm. Unlike two-stage approaches, YOLO models perform object localization and classification in a single inference pipeline, making them particularly attractive for applications where efficient processing is important. The YOLO family has evolved considerably over successive generations, introducing improvements to the network architecture, feature extraction, detection heads, training procedures, and loss functions. RetinaNet [7] is another single-stage detector that was proposed to address the problem of severe class imbalance between foreground and background samples during detector training. Its main contribution is the introduction of the Focal Loss function, which reduces the contribution of easily classified negative examples and focuses training on more difficult samples.

For the experiments presented in this work, object detec-

Name	Age filter	AP	AP@0.5	AP@0.75	AR	Time
YOLO3	none	0.511	0.654	0.552	0.739	0.07s
YOLO11x	none	0.478	0.584	0.508	0.708	0.08s
YOLO26x	none	**0.546**	**0.672**	**0.599**	**0.772**	0.05s
YOLO26x-E (open-world)	label=child	0.279	0.333	0.302	0.295	0.06s
FasterRCNN-50	none	0.364	0.568	0.378	0.582	0.08s
RetinaNet-50	none	0.324	0.523	0.321	0.524	0.08s
Cascade-MaskRCNN-50	none	0.383	0.542	0.405	0.612	0.14s

Table 1. Preliminary results for person detectors. Models are used without result filtering (no age estimation).

tors from two commonly used frameworks were selected. The first group consists of YOLO detectors implemented and distributed by Ultralytics [8][9]. Three YOLO versions were evaluated: YOLO3, YOLO11X, and YOLO26X. The second group consists of detectors available through the MMDetection toolbox [10], which provides implementations of a wide range of contemporary object detection architectures. Three models were selected from this framework: RetinaNet-50 [7], Faster R-CNN-50 [6], and Cascade Mask R-CNN-50 [11]. Since inference time and computational efficiency are not primary objectives of this experiment, large versions of models were chosen from the available architecture library.

The comparison also includes results obtained using an open-world YOLO model, which differs from the other evaluated detectors. Instead of relying on a fixed object category dictionary such as the COCO classes, open-world detection allows the user to define the categories of interest. In this experiment, two classes were specified: adult and child. From the perspective of the proposed task, this setup is particularly attractive because it could theoretically provide the desired child detections directly. Nevertheless, the obtained results were only moderate.

Table 1 presents the preliminary detection results. At this stage of the experiment, age estimation had not yet been introduced. Therefore, the person category predicted by the general-purpose detectors is treated as equivalent to the minor category in the ground-truth annotations. Predictions corresponding to all other COCO object categories are discarded before evaluation. For this preliminary detector comparison, recall is particularly important. The subsequent age estimation stage can only remove detections; it cannot recover a child that was not detected by the initial person detector. Consequently, errors at the first stage directly limit the maximum recall the complete pipeline can achieve. Among the evaluated general-purpose detectors, YOLO26x achieved the highest Average Recall (AR). It was therefore selected as the baseline detector for the subsequent analysis and for evaluating the combined person detection and age estimation pipeline.

5.1. Age estimation

Although age estimation is a well-established research topic in computer vision, it still presents challenges. One challenge is distinguishing apparent age from chronological age, as people may appear older or younger than they are. Another challenge, common in many human analysis tasks, is the high variability of representations: blurred and occluded faces and bodies, different poses, and poor lighting conditions. A particularly important limitation in the context of this work is that most methods rely primarily on facial information [12][13]. When the face is not visible or cannot be reliably detected, age estimation becomes considerably more difficult. Fewer methods attempt to estimate age from the entire body or other clues [3][14].

This work evaluates two age-estimation methods, with open-source code available to researchers. The first is InsightFace [12], which is a part of a larger face-analysis toolkit. Age estimation is performed on images where a face is visible and can be successfully detected. The age estimation component is implemented as an additional prediction head attached to a face recognition model based on the ResNet-50 architecture.

The second method is MiVOLO [14], which, in contrast, estimates age using both facial and body information. Its architecture consists of two branches, one processing the face and the other the body silhouette, with their representations fused within the network. The underlying architecture is based on Transformers, allowing information from both sources to contribute to the final age prediction.

Table 2 presents the detection results for the combination of the detector (YOLO26X) and selected age estimation methods. Detections for which the estimated age is greater than 14 are excluded from the final set. Detection patches for which no age estimation can be performed—for example, because a face cannot be detected—are classified as *unknown*. Bounding boxes classified as *unknown* are retained during evaluation. Evaluation metrics include average precision and recall, and the table also reports the number of image patches classified as adults, minors, and unknown. For comparison, the number of bounding boxes corresponding to children in the ground-truth annotations is provided.

Name	Age estimation	Evaluation					Age estimation results		
		AP	AP@0.5	AP@0.75	AR	Time	Adults	Minors	Unknown
YOLO26x	none	**0.546**	**0.672**	**0.599**	**0.772**	0.03s	-	-	-
YOLO26x	InsightFace (face)	0.223	0.263	0.245	0.302	0.21s	1492	8	1020
YOLO26x	MiVOLO (face+body)	0.428	0.500	0.451	0.454	0.08s	1367	1153	0
ground-truth bounding-boxes							-	1975	-

Table 2. Evaluation results for YOLO26X detector with selected age estimation methods. The table also includes class counts that were predicted for each patch in the dataset. The total number of detected bounding boxes was 2520.

The number is shown only for children, as the Parenting Slop Dataset does not annotate adults.

The age estimation experiments indicate that the selected methods struggle to reliably identify children in the analyzed images. For InsightFace, a large proportion of detection patches remains classified as unknown, primarily because the required facial information is unavailable or face detection fails. Among the patches for which an age estimate is produced, a considerable number of incorrect predictions can also be observed.

MiVOLO, which incorporates body information in addition to facial features, achieves better results than the face-based estimator. Nevertheless, its performance remains insufficient for reliable use as a filtering stage in the proposed multi-stage detection system. In both cases, age-based filtering yields lower precision than using the person detector without age filtering. As the libraries were not fully optimized, the average processing time is only approximate. In both cases, it was necessary to run an additional face detector or face-and-body detector that is part of the method to obtain the age value, so the pipeline redundantly applied detection steps.

These results suggest that age estimation methods should be improved and, in their current form, are not suitable for the proposed child detection pipeline. One solution is to develop a dedicated age estimation model adapted to the characteristics of the Parenting Slop Dataset, particularly images with small, occluded, or blurred faces, or subjects not looking into the camera. The second is to find more robust age estimation methods that can make reliable predictions from limited visual information.

6. MLLM-based child localization

In the previous experiment, classical person detectors combined with age-estimation methods were evaluated. SOTA person detectors achieved very high precision in the person-detection task. However, adding the available age-estimation methods generally decreased performance. Since training a dedicated age estimator is beyond the scope of this experiment, another class of methods that can potentially be used for child detection is investigated: multimodal large language models (MLLMs). These models can analyze both images and text, but they are not specifically optimized for object detection. In particular, it is difficult to consistently enforce a predefined output format, making their results harder to process automatically.

In the experiment described below, only open-weight models capable of running locally were considered. Five models from two providers, Google and Alibaba, were selected. Due to the limited time available for the experiment, not all models within each product line were evaluated. Table 3 presents the selected models and their parameters. The evaluated Google models were Gemma3-4B, a smaller model (4B parameters) and Gemma4-E4B, a larger and newer one (8B parameters). Three models from Alibaba were checked: the smallest one was Qwen3.5-4B (4B parameters), and Qwen3.6-27B, a larger and newer one (27B). Qwen3.6-35B-A3B was also tested, which, in contrast, is a mixture-of-experts model that has 35B parameters in total, but 3B are active when prompting. Only two models are comparable in size and capability: Gemma3-4B and Qwen3.5-4B. Other models differ substantially in scale and capacity. The selection criteria were mainly dictated by the technical constraints of running models, as these were the models that could be successfully deployed and evaluated on the available local hardware. All the models were downloaded from the Unsloth repository and deployed using Llama.cpp [15]. To make the generated responses as deterministic as possible, the temperature parameter was set to 0.0 for all evaluations.

For the detection task, the following prompt was used:

"Detect all the persons in the image and return their locations and descriptions in the form of coordinates. Results should be in JSON format: { 'user': {'bbox': [x1, y1, x2, y2], 'description': 'description of the person', 'label':'adult|child' } }"}".

For brevity, this prompt is referred to as the *all-people prompt*. It instructs the model to detect all persons in the image and assign each detection a label indicating whether the person is an adult or a child. For natural language models, the word *child* is used instead of *minor*, because it seems more universally understood.

A second prompt was also evaluated, explicitly restricting the task to children:

*"Detect all the **children** in the image and return their*

Name	Params	Input context	Weights	Correct JSONs	Fixable JSONs	Time
Qwen3.5-4B	4B	262K	2.9GB	12	931	32.52s
Qwen3.6-27B	27B	262K	17.6GB	31	999	117.04s
Qwen3.6-35B-A3B	35B (3B)	262K	22.4GB	10	999	13.73s
Gemma3-4B	4B	128K	2.5GB	0	1000	3.04s
Gemma4-E4B	8B	128K	5.1GB	571	993	4.30s

Table 3. Overview of MLLMs (multimodal large language models) used in the experiment

Method	Prompt	Postproc.	AP	AP@0.5	AP@0.75	AR
Gemma3-4b	all-people	label=child	0.000	0.001	0.000	0.003
Gemma4-E4B	all-people	label=child	0.153	0.314	0.135	0.294
Qwen3.5-4B	all-people	label=child	0.401	0.726	0.399	0.533
Qwen3.5-4B	children-only	none	0.417	0.733	0.413	0.567
Qwen3.6-27B	all-people	label=child	0.618	0.822	0.693	0.701
Qwen3.6-27B	children-only	none	**0.669**	**0.897**	**0.758**	**0.772**
Qwen3.6-35B-A3B	all-people	label=child	0.619	0.828	0.705	0.756
Qwen3.6-35B-A3B	children-only	none	0.585	0.829	0.645	0.734
YOLO26x	-	none	0.546	0.672	0.599	**0.772**

Table 4. Evaluation of child detection performed by querying MLLMs. Models were prompted to identify all people visible in the image with an additional label indicating whether each person is an adult or a minor (all-people prompt) or directly to find all children (children-only prompt). As Gemma models performed worse, experiments stopped after one type of prompt.

locations and descriptions in the form of coordinates. Results should be in JSON format: { 'user': {'bbox': [x1, y1, x2, y2], 'description': 'description of the person', 'label':'adult|child' } }. "

This prompt is referred to as the *children-only prompt*. Its purpose was to obtain child detections directly, without requiring an additional filtering step based on the predicted label.

Despite the detailed instructions, the models did not always return valid JSON. Even when the models provided correct detection boxes, outputs frequently contained formatting errors. Common problems included switching between bracket types or adding an extra level of JSON structure containing the *'user'* tag. The errors were also model-specific. For example, Qwen3.5-4B used both *'bbox'* and *'bbox_2d'* to denote the bounding-box coordinates. Qwen3.6-35B-A3B frequently produced problems with the 'label' attribute, filling it with *'person'* or *'persons'* instead of *'adult'* or *'minor'*, or omitting the attribute entirely.

Another issue was the coordinate system used by different models. Qwen models represent coordinates using a normalized 1000×1000 image. According to the documentation, Gemma also uses normalized coordinates, but its coordinate order during training differs: coordinates are represented as [y1, x1, y2, x2] instead of [x1, y1, x2, y2]. In practice, Gemma's results were considerably worse. One possible explanation is inconsistency between the coordinate normalization scheme and the format specified in the prompt. The models also differed in how they handled images containing no detectable persons. Gemma typically returned a natural-language response such as "I did not find any persons in the image", whereas Qwen generally returned a shorter message indicating that "no person was detected".

Overall, the outputs generated by the MLLMs required post-processing before they could be used for evaluation. Most formatting errors were relatively easy to correct, but both the type and frequency of errors depended on the model. Table 3 presents the number of responses that were considered correctly formatted for each model. A response was classified as correct if it contained an inner text section beginning with '[' or '' that could be successfully parsed as JSON, with a matching number of opening and closing brackets. The parsed structure also had to contain the required fields: *'bbox'* (or a similar bounding-box field), *'description'*, and *'label'*. Any additional text surrounding the JSON structure was ignored.

A response was classified as *fixable* if it contained valid coordinates and a description, and the description included a term indicating the person's age category. The terms considered sufficient for this purpose were *'person'*, *'adult'*, *'child'*, *'baby'*, *'boy'*, *'girl'*, *'man'*, and *'woman'*. Such responses could therefore be converted to the required format.

Table 4 presents the evaluation results for the five selected MLLMs. In the first evaluation setting, the models were instructed to localize all persons using the *all-people prompt*. The resulting detections were then filtered to retain only those assigned the *'child'* label. For the *all-people*

Name	Age estimation	Evaluation				Age estimation results		
		AP	AP@0.5	AP@0.75	AR	Adults	Minors	Unknown
YOLO26x	none	0.546	0.672	0.599	**0.772**	-	-	-
YOLO26x	Qwen3.5-4B	0.675	0.809	0.734	0.752	523	1955	42
YOLO26x	Qwen3.6-27B	0.682	0.810	0.743	0.734	666	1782	72
YOLO26x	Qwen3.6-35B-A3B	0.593	0.701	0.638	0.628	987	1508	25
ground-truth bounding-boxes						-	1975	-

Table 5. Evaluation results for YOLO26x detector with age estimation methods performed by MLLMs.

prompt, the highest precision was achieved by Qwen3.6-35B-A3B. However, its recall was slightly lower than that of the YOLO26X person detector, indicating it missed more relevant subjects. Gemma-family models did not perform well in the task of person detection; therefore, they weren't tested with the subsequent queries. Qwen-family models were additionally queried with a *children-only* prompt. For the children-only prompt, the results were highest for the model Qwen3.6-27B in terms of both average precision and recall. This model and prompting mode produced the highest overall score, but it was also the most time-consuming.

As shown in Table 4, the choice between directly prompting the model to find children and detecting all persons followed by filtering is not straightforward. For the model Qwen3.6-35B-A3B, filtering the results obtained with the *all-people prompt* produced higher precision. In contrast, Qwen3.6-27B achieved better results when using the *children-only prompt*. The difference cannot be clearly explained based on the current experiments, but the result demonstrates that MLLM-based detection can be sensitive to prompt formulation.

7. MLLM-based age estimation

In the experiments described in previous sections, classical object detectors were confronted with MLLMs. Classical detectors were fast and achieved high performance in person detection. However, age estimation became a bottleneck in the overall process, as age estimation errors could cause some of the detected bounding boxes to be incorrectly removed. MLLMs were able to perform the target task directly, namely detecting children in images. In addition, they provided valuable contextual information through descriptions of the persons visible in the image. However, the smallest models took about 100 times longer to infer than the baseline person detector. Moreover, the results require rule-based post-processing to fix the format inconsistencies.

As discussed earlier, several studies have proposed hybrid approaches that combine classical detectors with LLMs, aiming to exploit the strengths of both paradigms and enable zero-shot object detection capabilities. An additional experiment was therefore conducted to evaluate the combination of a classical person detector with an MLLM. In this setup, it replaced the conventional age-estimation component of the child-detection process.

The selected person detector, YOLO26X, generated the initial person detections used as input to the MLLM. For each detected person, the model received the corresponding image patch defined by the bounding box and was prompted to determine whether the person was a minor or an adult. The following prompt was evaluated:

"What age is the person in the image? The result should be either adult (for a person older than 14) or minor (for a person younger than 14). If there is no person in the image, return unknown."

The MLLM predictions were used in the same manner as the age estimation results in the previous experiment: patches classified as containing adults were filtered out, while the remaining detections were retained. The results obtained with the LLM-based age estimator are also presented in Table 5. It shows COCO evaluation results together with the number of patches classified as minor, adult, or unknown. The number of persons classified as minors was substantially higher than for conventional age estimators.

All evaluated Qwen models achieved higher evaluation results than both InsightFace and MiVOLO. Their precision was also higher than that of the baseline person detector without age-based filtering. Since age classification was performed only on patches generated by the baseline detector, recall could not exceed the baseline value. Nevertheless, Qwen3.6-27B and Qwen3.5-4B achieved recall values close to the baseline. Interestingly, the largest and smallest Qwen models produced similar performance results in this experiment. This approach was less time-consuming than end-to-end MLLM inference, likely because the patches were smaller than the full image. To sum up, combining the deterministic detection capabilities of classical detectors with the additional contextual capabilities of MLLMs provides a promising basis for more complex detection systems. Notably, the hybrid approach achieved higher AP and AP@75 than the end-to-end MLLM approach (all approaches are compared in Table 6).

Approach	Details	AP	AP@0.5	AP@0.75	AR	Time
Baseline detector	YOLO26x (no age est.)	0.546	0.672	0.599	**0.772**	0.05s
Detector + Age estimation with classification model	YOLO26x + MiVOLO	0.428	0.500	0.451	0.454	0.08s
End-to-end MLLM	Qwen3.6-27B (*children-only prompt*)	0.669	**0.897**	**0.758**	**0.772**	117.04s
Detector + Age estimation with MLLM	YOLO26x + Qwen3.6-27B	**0.682**	0.810	0.743	0.734	55.24s

Table 6. Summary of the selected approaches to child detection in the Parenting Slop Dataset.

8. Summary

This work compares three approaches to child detection built with two fundamentally different architectures: classical object detectors and multimodal large language models (MLLMs). All three approaches can be implemented with open-source models that can run offline on a local machine with a consumer-level graphics card, so they do not require sending sensitive data to an external server.

The final comparison in Table 6 summarizes results from all previous experiments. The first approach combines a person detector with an age-estimation model. Using YOLO26X as the baseline person detector and MiVOLO for age estimation resulted in an average precision (AP) of 0.428 and an average recall (AR) of 0.454. For comparison, the baseline person detector alone achieved an AR of 0.772 and found the most subjects in the dataset.

The second approach performs end-to-end child detection using the Qwen3.6-27B model, which is an example of an MLLM. This approach achieved significantly higher performance, with an AP of 0.669 and an AR of 0.772, but was very slow. However, its computational efficiency could potentially be improved.

The final approach combines both methods by using the person detector to localize people and the MLLM to estimate age (also with Qwen3.6-27B). The combined method achieved an AP of 0.682 and an AR of 0.734, producing results comparable to the end-to-end MLLM approach, with a higher AP, but lower AR.

Classical object detectors provide a deterministic and reproducible inference process, meaning that the same input produces the same output across repeated runs. Their outputs follow a predefined structure, and inference is generally fast. However, classical detectors are designed for specific tasks and data distributions, which limits their flexibility. When data characteristics or task requirements differ from those considered during model development, adapting the detector may require additional training or modifications to the underlying architecture. Consequently, the final performance depends strongly on the quality and representativeness of the data used for fine-tuning.

In this study, person detection was combined with age-estimation methods to filter adults from the detection results. However, this approach did not provide sufficient performance for the target task. Both tested age-estimation methods, MiVOLO and InsightFace, reduced the precision and recall compared with the baseline person detector without age-based filtering. One possible explanation is that these models rely heavily on facial information for age estimation, while many people in the dataset are captured under unconstrained conditions, with faces partially or completely invisible.

In contrast, multimodal large language models (MLLMs) offer greater flexibility and can perform complex visual reasoning and classification without task-specific training. This flexibility, however, comes at the cost of lower predictability and reproducibility. The generated output can vary depending on prompt formulation and may not follow the required format consistently. In particular, many responses were semantically close to the expected JSON structure but contained additional or malformed text that prevented direct JSON parsing, requiring heuristic post-processing. In addition, processing the entire dataset with an MLLM was considerably slower than using a classical object detector.

A third approach combines the advantages of both methods by first detecting persons and then using an MLLM to analyze each detected image patch. This approach achieved slightly higher average precision (AP) than end-to-end MLLM-based detection, but it also incurred a higher computational cost than the classical detectors. Despite these drawbacks, the combination of methods can provide a basis for extending the project into a more complex detection system. Its processing time could potentially be reduced through optimization, batching, or more efficient model deployment. Furthermore, the additional latency may be acceptable in applications where images are processed offline and real-time inference is not strictly required. A hybrid approach provides structured, reproducible outputs while extending detection models' capabilities with additional information from MLLMs that basic models lack without retraining.

9. Future work

The Parenting Slop Dataset consists of videos, but in this work, it is analyzed as a collection of static frames. Further improvements could be achieved by applying methods specifically designed for video data, such as object tracking, which could improve the results considerably. Video-based analysis could also be combined with MLLM-based methods.

The dataset is planned to be extended with additional video clips containing a higher proportion of adults or more challenging cases, such as older teenagers. This extension would allow a more thorough evaluation of age-estimation quality, since it is an important component of the overall detection pipeline.

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Table 7. Qualitative results of detection for the selected models

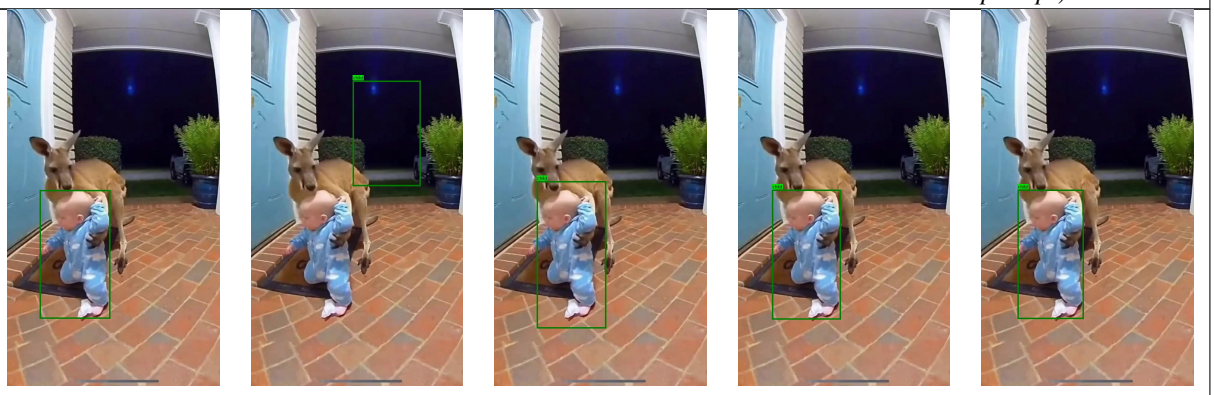
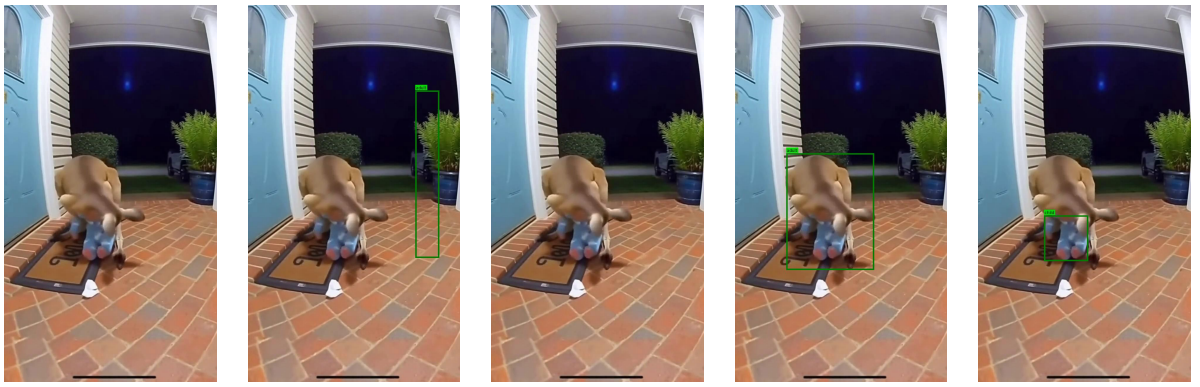
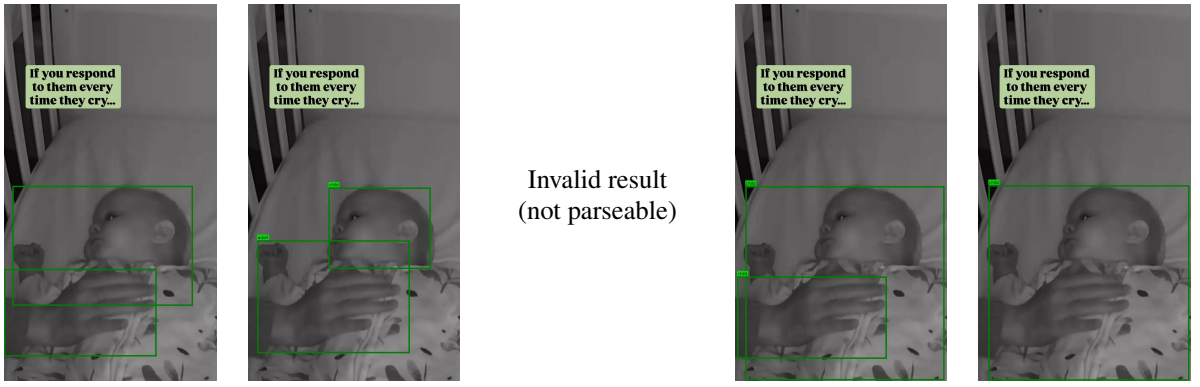
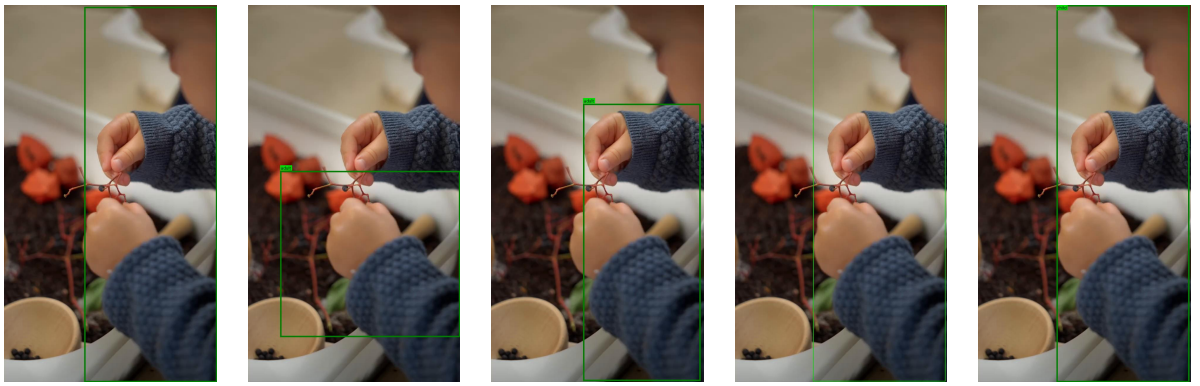
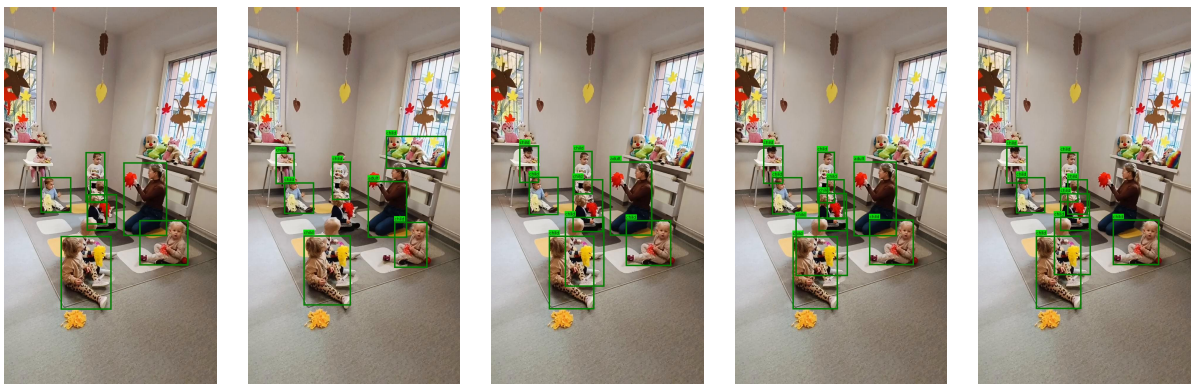
Frame ID	YOLO26x	Gemma4-E4B (*all-people prompt*)	Qwen3.5-4B (*all-people prompt*)	Qwen3.6-35B-A3B (*all-people prompt*)	Qwen3.6-27B (*children-only prompt*)
06/000					
06/066					
00/000			Invalid result (not parseable)		
08/000					

Table 7. Qualitative results of detection for the selected models

Frame ID	YOLO26x	Gemma4-E4B (*all-people prompt*)	Qwen3.5-4B (*all-people prompt*)	Qwen3.6-35B-A3B (*all-people prompt*)	Qwen3.6-27B (*children-only prompt*)
11/108					
13/080					
14/000					
18/096	